



Nitrogen recovery in planted seedlings, competing vegetation, and soil in response to fertilization on a boreal mine reclamation site



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ABSTRACT

Field fertilization during reforestation often yields variable results, particularly on harsh restoration sites. An improved understanding of the recovery of applied nitrogen (N) under different fertilization practices should aid in developing more effective fertilizer prescriptions. We evaluated field establishment of white spruce (*Picea glauca* (Moench) Voss) and trembling aspen (*Populus tremuloides* Michx.) seedlings as well as N recovery within planted seedlings, soil, and competing vegetation on a mine reclamation site in the oil sands region of northern Alberta in response to immediately available fertilizer (IAF) and polymer-coated controlled-release fertilizer (CRF) applications. ¹⁵N-enriched urea was applied as IAF and as a polymer-coated CRF (20 g N seedling⁻¹ and 4 g N seedling⁻¹, respectively) to each species. Seedling survival, growth, and nutritional status, along with occurrence of competing vegetation and plant and soil ¹⁵N recovery were quantified after the first field season. Seedlings receiving CRF exhibited increased diameter and organ dry mass relative to the IAF and control treatments. Both IAF and CRF promoted comparable increases in seedling N status, and fertilizer type did not influence within-seedling ¹⁵N allocation. Neither IAF nor CRF affected vegetation cover or dry mass. Recovery of fertilizer-derived ¹⁵N was low, with much of the recovered ¹⁵N remaining in soils and only small amounts observed in seedlings and competing vegetation for both fertilizer treatments. Findings indicate that directed root zone application of CRF promotes first-year seedling growth and nutritional responses similar to or better than those induced by broadcast IAF applications, but at substantially lower N application rates. Our results suggest that a shift from broadcast IAF to targeted soil applications of CRF may produce similar or improved early seedling growth and nutrient uptake on reclamation sites, while greatly reducing overall quantities of N applied during the regeneration phase, much of which appears to be lost from the site of application regardless of fertilizer type.

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1. Introduction

Surface mining of oil sands deposits has disturbed over 715 km² of boreal forest in northeastern Alberta (The Oil Sands Developers Group, 2009; Alberta Government, 2014). After mining takes place, reclamation of the affected area is required, with the goal of achieving a level of ecological capability equivalent to that which existed prior to disturbance (Alberta Environment, 2006). Thus, reclaimed areas in the oil sands region are required to have species characteristic of native plant communities (Alberta Environment, 2009), but extensive reclamation efforts are required to return these lands to productive forest lands that meet needs for

ecosystem services (Ciccarese et al., 2012; Oliet and Jacobs, 2012; Jacobs et al., 2015).

Soils in NE Alberta's reclamation areas are often reconstructed using a mixture of peat and mineral soil materials that were salvaged prior to mining (Sorenson et al., 2011; MacDonald et al., 2015). These new soils may have low fertility as well as substantial physical and chemical differences compared with natural soils, such as high soil pH, increased salinity, and compaction in the overburden (Farnden et al., 2013; Lilles et al., 2012; Sorenson et al., 2011). These areas are then planted with species such as trembling aspen (*Populus tremuloides* Michx.), jack pine (*Pinus banksiana* Lamb.), and white spruce (*Picea glauca* (Moench) Voss), which are components of local native plant communities and representative of the pre-mining species composition (Fung and Macyk, 2000; Johnson et al., 1995), as well as relatively fast growing and drought tolerant (Liefers et al., 2001). These species have also shown tolerance to salinity and other limitations that may

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characterize soils on reclaimed mine sites (Lieffers et al., 2001; Lilles et al., 2012). Seedlings of these species often exhibit transplant stress, however, and grow slowly for several years after outplanting (Martens et al., 2007; van den Driessche et al., 2003).

While site limiting factors vary across regions, attempts to improve regeneration success on mine reclamation sites have included herbicide use to control competing vegetation and field fertilization (Andersen et al., 1989; Casselman et al., 2006). Rapid response to weed control has been seen for other fast-growing tree species such as loblolly pine (*Pinus taeda* L.) (Perry et al., 1993) and *Populus* spp. (Pinno and Bélanger, 2009), but slower growing species with determinate growth patterns, such as Engelmann spruce (*Picea engelmannii* Parry ex Engelm.), may require up to three years after repeated competition control to show any growth response (Biring et al., 2003). Despite the benefits of herbicide application for reducing vegetative competition, there is increasing public sentiment against its use due to negative effects on the environment and on biodiversity (Löf et al., 2012; Thiffault and Roy, 2010).

Fertilization at the time of planting has also been shown to increase early growth rates, allowing trees to overtop competing vegetation and accelerate stand establishment (Miller, 1981; Rowland et al., 2000). Current reclamation practices include applying various fertilizer prescriptions based on perceived needs for nutritional supplements, frequently with broadcast applications of traditional (agronomic), immediately available fertilizers (IAF) for up to 5 years after planting (MacKenzie, 2011; Pinno et al., 2012). However, IAF has generally shown low recovery rates and may stimulate growth and nutrient uptake of competing vegetation more than outplanted seedlings (Chang and Preston, 2000; Chang et al., 1996; Imo and Timmer, 1998; Ramsey et al., 2003; Staples et al., 1999). Excessive nutrient supply, commonly resulting from an application of conventional, water-soluble fertilizers, may result in a high concentration of soluble salts in the root zone (Shaviv and Mikkelsen, 1993; Trenkel, 1997). Furthermore, detection of relatively high or low nitrate levels in some fertilized stands in the oil sands region (Rowland et al., 2000) suggests the need to optimize fertilization operations to improve cost efficiency and decrease the risk of nutrient leaching to surface and underground water bodies (McMillan et al., 2007).

Steady-state nutrition theory (Ingstad and Lund, 1986) suggests that seedling growth and nutrient uptake can be maximized and leaching losses minimized by supplying nutrient quantities in proportion to plant requirements. Controlled-release fertilizers (CRF) allow nutrients to be released slowly over time to better match plant demand and are expected to improve use efficiency (UE) and minimize adverse effects on the environment (Donald, 1991; Shaviv, 2005). Polymer-coated CRFs provide especially good control over release rates because they tend to be less sensitive to soil conditions (Shaviv, 2005). This allows for directed application of fertilizer to the seedling root zone at planting, with relatively low risk of root damage compared to IAF (Jacobs and Timmer, 2005). A single application of CRF can maintain nutrient availability at the levels of plant demand over an extended period of time, as they can release nutrients for up to two years (Fan et al., 2002; Hangs et al., 2003; Jacobs et al., 2005). Controlled-release fertilizer has been shown to improve plant growth and quality, increase nutrient use efficiency, reduce the cost of maintenance associated with repeated fertilizations, and decrease nutrient losses to surface water (Hangs et al., 2003; Shaviv and Mikkelsen, 1993; Shaviv, 2001). Nutrient release of CRF begins when a critical volume of saturated solution is formed inside the fertilizer prill. Thus, when fertilization occurs at the time of outplanting in late winter or spring, nutrient release generally occurs after the onset of competing vegetation, reducing its negative effect on seedling performance (Shaviv, 2005). Although there are different methods of applying CRF, application directly (or immediately adjacent) to the seedling

root zone at planting has relatively low risk of root damage compared to IAF (Jacobs and Timmer, 2005). Application of CRF at planting has stimulated growth of newly planted forest trees across a variety of ecotypes in North America (e.g., Arnott and Burdett, 1988; Carlson and Preisig, 1981; Carlson, 1981; Fan et al., 2002; Jacobs et al., 2005).

Despite the demonstrated potential to improve UE of fertilizer operations on reforestation and afforestation sites, most studies investigating the use of CRF in field plantings have not directly compared CRF and IAF (and their corresponding application methods) in terms of seedling development and nutrient uptake dynamics. Additionally, while there are many examples of applied fertilizer studies in forest regeneration, relatively few of these have assessed the recovery of N for varying fertilization methods using ^{15}N tracing techniques. Past fertilizer recovery studies have also generally been conducted on reforestation sites following logging, while our study was focused on reclamation of a heavily disturbed post-mining site, which represents an important new ecosystem for this type of research given the increasing significance of these sites as restoration targets (MacDonald et al., 2015).

Sloan and Jacobs (2013) recently examined seedling responses to a wide range of CRF and IAF rates on a boreal post-mining reclamation site, reporting improved seedling growth for both fertilizer types relative to controls. They noted that responses to CRF application were similar or better to those achieved using IAF, despite using 90–95% lower N application rates. These results, implying wide discrepancies in UE between the two fertilizer systems, indicate the need for more detailed examination of relative rates of recovery of fertilizer N within planted seedlings or in adjacent competing vegetation and soil. To help address these knowledge gaps, we conducted a field trial on a mine reclamation site in the Canadian oil sands region to examine use of CRF in comparison to IAF. Our objectives were to: (1) characterize and quantify the recovery of ^{15}N -enriched fertilizer during the first post-planting growing season using different application methods (i.e. planting hole fertilization for CRF and broadcast for IAF); (2) examine responses of competing vegetation among fertilizer treatments; and (3) compare seedling growth and nutrient uptake between CRF and IAF.

2. Materials and methods

2.1. Study site and plantation establishment

The study site was located on a level portion of a reclamation area of an oil sands mine located north of Fort McMurray, Alberta, Canada (56°56.495'N, 111°16.443'W, 369 m ASL). Substrate on the site consisted of a peat-mineral mix (PMM) capping material (excavated from low-lying peatland sites, in which the layer of organic soil and some underlying mineral soil were salvaged) averaging 50 cm in depth placed atop an overburden substrate. Soil analysis of this PMM material from adjacent plots on the same mine reclamation site showed 52.3% sand, 14.2% clay, 6.67 pH, and electrical conductivity of 0.125 dS m⁻¹ (Schott et al., 2015). Historical mean summer/winter (May–August/September–April) air temperatures in the area are 14.3/−6.0 °C and seasonal precipitation is 265.7/190.0 mm (Environment Canada: National Climate Data and Information Archive 2013).

On 9 June 2011, 30 seedlings per species of white spruce (*P. glauca* (Moench) Voss) and trembling aspen (*P. tremuloides* Michx.) were randomly arranged and planted at 3 m × 3 m spacing. All seedlings of both species were grown in 615A styroblock containers (336 cm³ volume, 15.1 cm depth × 5.9 cm top diameter, Beaver Plastics, Ltd., Acheson, Alberta, Canada) and produced operationally at the Smoky Lake Forest Nursery of Coast to Coast Refor-

estation, Inc. (Smoky Lake, Alberta, Canada). Within each species, each seedling was randomly assigned to one of three fertilizer treatments at time of planting.

2.2. Fertilization

Fertilizer treatments were applied the day after planting and consisted of an unfertilized treatment (control), a ^{15}N -enriched controlled-release fertilizer treatment (CRF), or a ^{15}N -enriched immediately available fertilizer treatment (IAF). Control seedlings received no fertilizer application at time of planting and served as an unlabeled isotopic control. Seedlings assigned to the CRF treatment received 4 g N seedling⁻¹ dibbled into the soil at a depth of 5 cm at a distance of 10 cm from the base of the seedling stem applied as 9 g ^{15}N -enriched ESN[®] (Environmentally Smart Nitrogen) polymer-coated urea (custom-manufactured by Agrium U.S. Inc., Denver, CO, USA using 0.5 atom% ^{15}N -enriched urea custom-manufactured by Applied Chemical Technology Inc., Florence, AL, USA). Seedlings assigned to the IAF treatment received 20 g N seedling⁻¹ broadcast on the soil surface within a 0.5 m² area surrounding the seedling (equivalent to a field-level application rate of 400 kg N ha⁻¹) applied as 43 g ^{15}N -enriched urea (0.5 atom% ^{15}N , custom-manufactured by Applied Chemical Technology Inc., Florence, AL, USA). Because fertilizer use efficiency of CRF is expected to be greater than that of IAF, fertilizer rates for CRF may be considerably lower than those used operationally for IAF. For both CRF and IAF, these N rates produced positive nutrient uptake and growth responses over unfertilized seedlings in another fertilizer study on a boreal mine reclamation site (Sloan and Jacobs, 2013).

2.3. Measurements and sampling

Initial total height and root collar diameter were measured for each seedling at time of planting; no competing vegetation was present at that time. On 10 September 2011, at the end of the first growing season, survival, final total height, and root collar diameter were assessed for each seedling, and competing vegetation cover was measured as % cover within a 0.5 m² sampling grid centered on each seedling. A 2 cm × 15 cm (diameter × depth) soil core was extracted from four points surrounding each seedling (to the northeast, southeast, southwest, and northwest) at a lateral distance of 15 cm from the root collar. The four soil cores from each seedling were then mixed and homogenized to form a composite sample.

On 11 September 2011, the aboveground portion of all competing vegetation emerging from within each seedling's 0.5 m² sampling grid was harvested. Additionally, all seedlings were carefully excavated and harvested by hand, roots were washed free of soil, and seedlings were separated into organs (roots, stems, and foliage). Thus, data collection was limited to the first growing season, similar to previous reforestation experiments with ^{15}N (Clinton and Mead, 1994; Hanks et al., 2003); both fertilizer types were designed to release nutrients within the first season of application and the ability to detect recovery of ^{15}N in seedlings, competing vegetation, and soil is therefore expected to diminish in subsequent years.

Seedling organs and competing vegetation samples were dried at 68 °C to a constant mass and dry mass of each sample was recorded, after which all samples were ground in a Wiley mill (Thomas Scientific, Swedesboro, NJ, USA) to pass a 20-mesh screen. Soil samples were air-dried to constant mass at 23 °C, after which the dry mass of each sample was recorded before being hand-ground to pass a 20-mesh screen using a mortar and pestle. All samples were then submitted to the Stable Isotope Core Laboratory of Washington State University for determination of N concentra-

tion and ^{15}N abundance via isotope ratio mass spectrometry (Stable Isotope Core, Washington State University, Pullman, WA, USA). Seedling foliage samples were also submitted to A & L Great Lakes Laboratories, Inc. (Fort Wayne, IN, USA) for analysis of the following macro- and micro-nutrients: P (%), K (%), Ca (%), Mg (%), S (%), Zn (mg kg⁻¹), Mn (mg kg⁻¹), Fe (mg kg⁻¹), Cu (mg kg⁻¹), B (mg kg⁻¹), Al (mg kg⁻¹), and Na (%).

2.4. Calculations and statistical analysis

First-season height growth was calculated by subtracting seedling initial height from final height. First-season diameter growth was calculated similarly. Nitrogen concentration of seedling organs reflects % N values reported on a dry mass basis. Nitrogen content was determined by multiplying the measured % N of a sample by its measured dry mass to yield a total N content for each organ of each seedling. Seedling ^{15}N content for each organ reflects the total ^{15}N content, uncorrected for ^{15}N natural abundance. For the % assimilated ^{15}N values, seedling organ ^{15}N contents were corrected for natural abundance of ^{15}N based upon mean values observed in the control seedlings so as to estimate the fertilizer-derived ^{15}N content of each seedling organ. The resulting corrected values were then expressed as % assimilated ^{15}N , relative to the total quantity of assimilated fertilizer-derived ^{15}N on a dry mass basis. For fertilizer fate, values of % applied ^{15}N are based upon total quantities of fertilizer-derived ^{15}N observed in each identified sink (seedling, competing vegetation, or soil) following correction for natural abundance based on mean values of corresponding samples from controls, expressed relative to the total quantity of ^{15}N applied as fertilizer. Recovery of ^{15}N reflects the total quantity of fertilizer-derived ^{15}N accounted for through the described sampling methods at the end of the first growing season, relative to quantities of ^{15}N applied as fertilizer.

The experiment was established and analyzed as a completely randomized factorial design. The independent variables species and fertilizer treatment constituted the factors for the dependent variables of seedling survival, final height, height growth, final diameter, diameter growth, occurrence of competing vegetation, and dry mass of competing vegetation. The independent variables species, fertilizer treatment, and seedling organ constituted the factors for the dependent variables of biomass allocation, N concentration, N content, and ^{15}N content of seedling organs, as well as for within-seedling allocation of assimilated ^{15}N . The independent variables species, fertilizer treatment, and sink constituted the factors for analysis of fertilizer fate, while the independent variables species and fertilizer treatment constituted the factors for analysis of overall recovery of applied ^{15}N . All data were analyzed using ANOVA to detect main effect significance ($P < 0.05$) followed by Tukey's multiple pairwise comparison to detect significant differences between treatment means ($\alpha = 0.05$). All data analyses were conducted using SAS software version 9.4 (SAS Institute, Inc., Cary, NC, USA).

3. Results

3.1. Seedling survival, growth, biomass allocation, and vegetative competition

Seedling survival was not influenced by species or fertilizer treatment (Table 1; data not shown) and was close to 100% in all treatments for both species. Additionally, although initial and final height were influenced by species ($P < 0.01$ and $P = 0.03$, respectively; data not shown), with trembling aspen exhibiting greater final heights than white spruce, fertilization did not influence first season final height ($P = 0.11$; data not shown). No significant

Table 1

ANOVA results for seedling survival and growth, competing vegetation, seedling N dynamics, and fertilizer fate. For each entry, *P*-values are displayed for the overall model, main effects, and all main effect interactions. Values in bold were significant at $P < 0.05$.

Dependent variable	Model	Species	Fert	Species × fert
Survival	0.4110	0.0500	0.7497	0.7491
Final height	0.0214	0.0318	0.1103	0.0923
Height growth	0.2163	0.1786	0.1418	0.3635
Final diameter	<0.0001	<0.0001	0.0064	0.2510
Diameter growth	<0.0001	<0.0001	0.0090	0.6156
Occurrence of vegetation	0.6288	0.6057	0.3773	0.6026
Dry mass of vegetation	0.3753	0.7503	0.5065	0.1551
Recovery of applied ^{15}N	0.0005	0.4553	<0.0001	0.9339

Dependent variable	Model	Species	Fert	Organ	Species × fert × organ	Species × fert	Species × organ	Fert × organ
Seedling biomass allocation	<0.0001	<0.0001	0.0019	0.0087	0.6973	0.1392	<0.0001	0.6710
Seedling N concentration	<0.0001	<0.0001	<0.0001	<0.0001	0.8772	0.9397	<0.0001	<0.0001
Seedling N content	<0.0001	<0.0001	<0.0001	<0.0001	0.1848	0.5578	<0.0001	0.0046
Seedling ^{15}N content	<0.0001	<0.0001	<0.0001	<0.0001	0.2051	0.5423	<0.0001	0.0033
Within-seedling ^{15}N allocation	<0.0001	0.8044	0.8044	<0.0001	0.4082	0.8044	<0.0001	0.3471

Dependent variable	Model	Species	Fert	Sink	Species × fert × sink	Species × fert	Species × sink	Fert × sink
Fertilizer fate	<0.0001	0.4637	<0.0001	<0.0001	0.9493	0.9278	0.7276	<0.0001

fertilizer or species effects were observed for first season height growth ($P = 0.22$; data not shown). However, despite no significant differences in initial diameter between species ($P = 0.31$; data not shown), both species and fertilizer treatment affected first season final diameter (Table 1; data not shown) and first season diameter growth (Table 1; data not shown), with white spruce seedlings exhibiting larger final diameters and increased diameter growth relative to trembling aspen and seedlings that had received CRF out-performing unfertilized seedlings, but not differing from IAF treatments. No species × fertilizer treatment interactions were observed for the above-mentioned parameters.

With regard to seedling biomass allocation, both fertilizer treatment (Table 1; data not shown) and a species × organ interaction influenced seedling organ biomass (Table 1; Fig. 1). Average organ biomass of seedlings that had received CRF increased significantly compared with those that received either no fertilizer or IAF. Biomass allocation differed markedly between white spruce and trembling aspen seedlings, with biomass of white spruce seedlings concentrated in stems and foliage, while that of trembling aspen seedlings was found primarily in roots and stems (Fig. 1).

Occurrence (i.e. % cover) and dry mass of competing vegetation surrounding seedlings were unaffected by seedling species or fertilizer treatment (Table 1; data not shown).

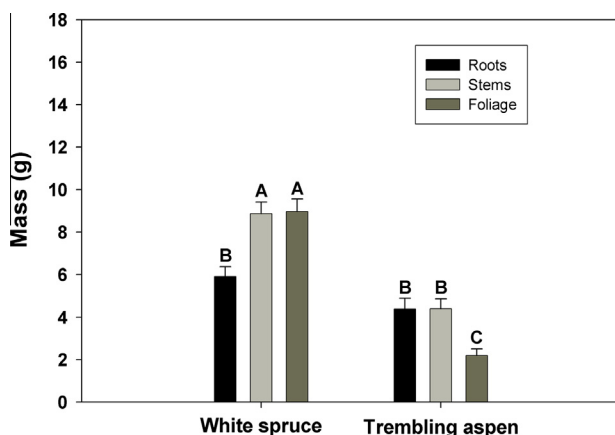


Fig. 1. Seedling biomass allocation (g) for roots, stems, and foliage of white spruce and trembling aspen seedlings at the end of the first growing season. Values displayed are means ± SEM and columns labeled with the same letter do not differ significantly at $\alpha = 0.05$.

3.2. Seedling nitrogen dynamics and foliar nutrition

Highly significant seedling organ × fertilizer treatment interactions were observed for N concentration, N content, and ^{15}N content of seedling organs (Table 1; Fig. 2). Both CRF and IAF treatments resulted in similar increases in foliar and root N concentrations in foliage and roots of fertilized seedlings relative to those of unfertilized seedlings (Fig. 2A). Likewise, the total foliar N content of both CRF and IAF seedlings exceeded that of the unfertilized seedlings, and the N content of the roots of CRF seedlings exceeded that observed in roots of unfertilized control seedlings (Fig. 2B). Neither N concentration nor N content varied in stems in response to fertilizer treatment (Fig. 2A and B). With the exception of stems in the IAF treatment, ^{15}N content of all organs in fertilized seedlings was highly enriched relative to that of the corresponding organs in unfertilized control seedlings and closely mirrored the patterns observed for N concentration and content (Fig. 2C).

Highly significant species × seedling organ interactions were also observed for N concentration, N content, and ^{15}N content of seedling organs (Table 1; Fig. 3A–C, respectively). Nitrogen concentration of both species was highest in foliage, followed by roots and stems; foliar N concentrations of trembling aspen greatly exceeded all other observed seedling organ N concentrations (Fig. 3A). However, white spruce foliage exhibited the greatest N content and ^{15}N content, with no other differences observed for N or ^{15}N content across other organs (Fig. 3B and C).

Within-seedling allocation of assimilated ^{15}N was unaffected by fertilizer treatment (Table 1; data not shown), but a highly significant species × seedling organ interaction was observed (Table 1; Fig. 3D). White spruce seedlings were shown to allocate the vast majority of assimilated ^{15}N to foliage, with much lower and roughly equal amounts being allocated to stems and roots; trembling aspen, on the other hand, allocated roughly equal amounts of assimilated ^{15}N to both foliage and roots, with a somewhat lower amount being allocated to stems (Fig. 3D).

With regard to other foliar nutrients, trembling aspen had significantly higher foliar concentrations of K, Ca, Mg, S, Mn, Cu, and B ($P < 0.01$), compared with white spruce that had higher concentrations of Fe and Al (Table 2). Foliar Ca and B were reduced in the IAF treatment relative to control seedlings ($P = 0.03$ and <0.01 , respectively; Table 2). A species × fertilizer interaction was observed for foliar concentrations of P and Zn ($P = 0.01$ and $P < 0.01$, respectively; Table 2).

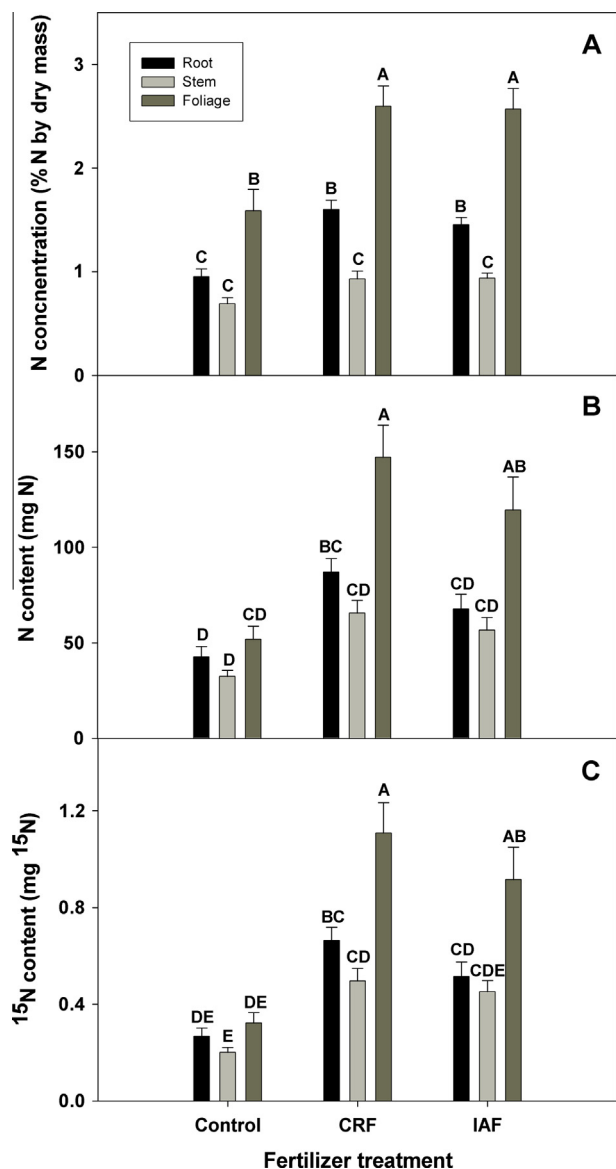


Fig. 2. Nitrogen concentration (% N by dry mass; A), content (mg N; B), and ^{15}N content (mg ^{15}N ; C) of roots, stems, and foliage of unfertilized control seedlings (Control), seedlings that received 4 g N seedling $^{-1}$ via 0.5 atom% ^{15}N -enriched polymer-coated controlled-release urea applied at time of planting (CRF), and seedlings that received 20 g N seedling $^{-1}$ via 0.5 atom% ^{15}N -enriched immediately available urea applied at time of planting (IAF). Values displayed are means (averaged across both white spruce and trembling aspen seedlings) $\pm$ SEM and columns labeled with the same letter do not differ significantly at $\alpha = 0.05$.

3.3. ^{15}N Nitrogen recovery

Although species did not influence recovery of applied fertilizer N, a highly significant sink by fertilizer treatment interaction occurred (Table 1; Fig. 4). At the end of the first growing season, the soils surrounding IAF and, to a lesser extent, CRF seedlings had the greatest concentration of applied ^{15}N compared with seedlings and vegetation (Fig. 4).

Overall recovery of applied ^{15}N was low and varied significantly by fertilizer treatment (Table 1; Fig. 4), with a much greater proportion of applied ^{15}N being recovered from the IAF treatment (13.78%) compared with the CRF treatment (6.46%), reflecting 86.22% and 93.54% of applied ^{15}N being unaccounted for after one growing season for the IAF and CRF treatments, respectively. Despite the lower ^{15}N recovery observed in the CRF treatment, a

much lower total quantity of ^{15}N remained unaccounted for in the CRF treatment (3.74 g N seedling $^{-1}$ unaccounted for) compared to the IAF treatment (17.24 g N seedling $^{-1}$ unaccounted for) due to the different application rates associated with these products. Tree species was not observed to influence overall ^{15}N recovery (Table 1).

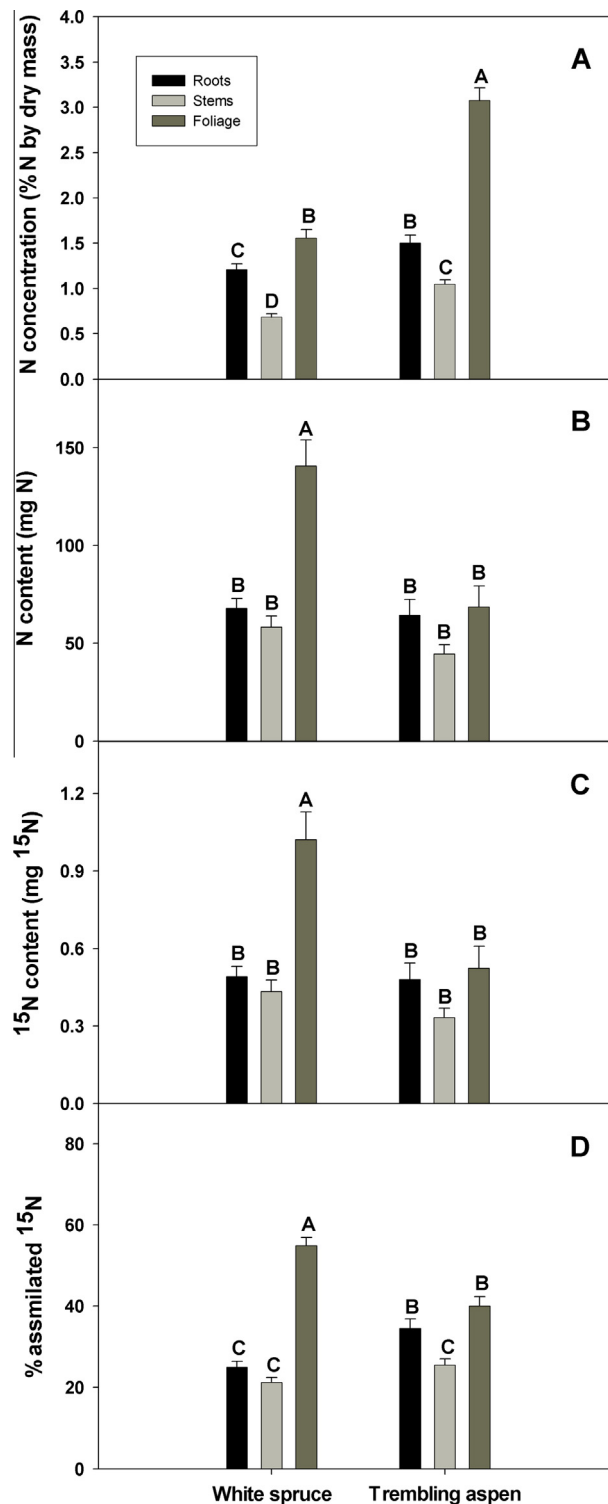
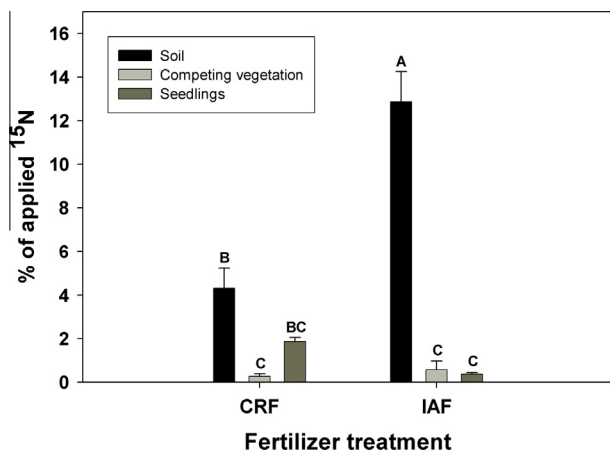


Fig. 3. Nitrogen concentration (% N by dry mass; A), content (mg N; B), ^{15}N content (mg ^{15}N ; C), and allocation of assimilated ^{15}N (%; D) for roots, stems, and foliage of white spruce and trembling aspen seedlings at the end of the first growing season. Values displayed are means $\pm$ SEM and columns labeled with the same letter do not differ significantly at $\alpha = 0.05$.

Table 2Foliar nutrient concentrations by species and fertilizer treatment. Values displayed are means ($\pm$ SEM).

	White spruce (<i>Picea glauca</i>)			Trembling aspen (<i>Populus tremuloides</i>)		
	Control	IAF	CRF	Control	IAF	CRF
P (%)	0.10 (0.00) BC	0.08 (0.00) BC	0.07 (0.00) C	0.19 (0.02) A	0.12 (0.01) B	0.11 (0.01) BC
K (%) [*]	0.60 (0.02)	0.37 (0.02)	0.35 (0.02)	0.83 (0.06)	0.88 (0.11)	0.91 (0.10)
Ca (%) ^{*,†}	0.93 (0.05)	0.88 (0.06)	0.90 (0.05)	2.40 (0.20)	1.76 (0.16)	2.14 (0.17)
Mg (%) [*]	0.17 (0.01)	0.16 (0.01)	0.19 (0.01)	0.59 (0.04)	0.50 (0.04)	0.55 (0.03)
S (%) [*]	0.16 (0.01)	0.12 (0.00)	0.12 (0.00)	0.35 (0.02)	0.36 (0.04)	0.32 (0.02)
Zn (mg kg ⁻¹)	21.86 (1.57) C	16.67 (2.44) C	17.44 (1.32) C	497.26 (83.61) A	107.95 (22.91) C	284.56 (61.55) B
Mn (mg kg ⁻¹) [*]	113.53 (15.06)	130.73 (13.22)	106.25 (11.80)	186.97 (20.52)	157.92 (13.56)	122.65 (12.04)
Fe (mg kg ⁻¹) [*]	349.21 (18.58)	339.89 (25.58)	320.02 (21.18)	243.34 (17.37)	258.27 (18.64)	278.28 (11.31)
Cu (mg kg ⁻¹) [*]	1.47 (0.21)	1.65 (0.16)	1.73 (0.17)	5.67 (0.35)	4.86 (0.51)	5.85 (0.54)
B (mg kg ⁻¹) ^{*,†}	19.35 (1.41)	10.96 (0.59)	13.96 (0.99)	47.83 (4.95)	33.32 (3.46)	38.63 (4.02)
Al (mg kg ⁻¹) [*]	183.10 (10.49)	181.22 (15.62)	187.30 (12.64)	130.00 (10.02)	109.4 (11.33)	117.67 (7.30)
Na (%)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.02 (0.00)

Where a species $\times$ fertilizer interaction occurred, means followed by the same letter within a row did not differ significantly ($\alpha = 0.05$).^{*} Significant species effect.[†] Significant fertilizer treatment effect.**Fig. 4.** Proportion of total applied ^{15}N recovered from soil, competing vegetation, and seedlings after the first growing season for seedlings which received 4 g N seedling⁻¹ via 0.5 atom% ^{15}N -enriched polymer-coated controlled-release urea applied at time of planting (CRF), and for seedlings which received 20 g N seedling⁻¹ via 0.5 atom% ^{15}N -enriched immediately available urea applied at time of planting (IAF). Values displayed are means $\pm$ SEM and columns labeled with the same letter do not differ significantly at $\alpha = 0.05$.

4. Discussion

4.1. Differential responses by species

White spruce seedling mass was comprised mainly of needles and stems, with less biomass allocated to roots whereas trembling aspen leaf biomass was lower relative to stems and roots (Fig. 1). These distinct patterns of biomass allocation were also reflected in species-specific patterns of N content (Fig. 3A and B). Additionally, ^{15}N allocation to white spruce needles was significantly higher than to other organs, whereas trembling aspen allocated newly acquired ^{15}N more proportionally to all organs (Fig. 3C and D). Differences between species can be attributed to the leaf habit and ecology of the species. Deciduous species tend to store N in woody organs (Millard and Proe, 1991; Millard, 1996; Silla and Escudero, 2003; Villar-Salvador et al., 2015) while needles, especially young ones, are major N storage sites in evergreen species (Nambiar and Fife, 1991; Silla and Escudero, 2003; Villar-Salvador et al., 2015). Other studies have shown that the root system of trembling aspen is typically deeper and contains a higher proportion of plant biomass than that of white spruce, as was observed in this study (Gale and Grigal, 1987; Strong and La Roi, 1983). The higher ^{15}N

in foliage of both species and in roots of trembling aspen reflects the relative sink strength of these organs (Fig. 3D). Newly developing organs are typically the main sinks for resource allocation within the plant, and new tissues, especially shoots and roots, act as strong resource sinks during growth (Cerasoli et al., 2004; Chapin III et al., 1990; Hansen et al., 1996; Nambiar and Fife, 1991; Uscola et al., 2015).

4.2. Effects of fertilization on planted seedlings

Both fertilization treatments increased N concentration and content of seedlings compared to the controls (Fig. 2). Although both IAF and CRF promoted seedling N status, seedlings that received CRF had increased diameter growth and final diameter compared to controls, as well as greater seedling organ mass and N content of foliage and roots relative to both unfertilized controls and IAF seedlings. This was despite the 80% lower rate of N applied to seedlings in the CRF treatment. Several experiments have reported improved performance of seedlings fertilized with CRF compared to IAF, in both controlled and field experiments (Carlson and Preisig, 1981; Fan et al., 2002; Haase et al., 2006; Hangs et al., 2002; Sloan and Jacobs, 2013). This can be explained firstly by the slower release rate of N by the CRF, which allows the N to become available over a longer period of time and at a rate more commensurate with patterns of seedling N demand (Donald, 1991; Ingestad and Lund, 1986; Shaviv, 2001). Secondly, the placement of CRF directly in the root zone provides roots, and especially new roots, with immediate and continuous access to nutrients (Cabrera, 1997; Hangs et al., 2003; Jacobs and Timmer, 2005), and CRF fertilizer placed directly in the planting hole has been shown to result in greater seedling growth than placement in an adjacent hole for both Douglas-fir and western hemlock (Carlson and Preisig, 1981; Carlson, 1981).

Survival of both species was close to 100% in all treatments indicating that nutrient ion concentrations and electrical conductivity levels fell outside the range of toxicity (Jacobs and Timmer, 2005). Lower growth responses to fertilizer application in this study relative to other field fertilization studies (DesRochers et al., 2006; Carlson and Preisig, 1981; Fan et al., 2002) can be attributed to the tendency for low first year growth for seedlings planted on reclaimed lands (Chaney et al., 1995; Martens et al., 2007; van den Driessche et al., 2003), with fertilization effects being usually more evident in diameter than in height growth (Fan et al., 2002). Additionally, growth has been shown to depend more on N reserves from previous years than on N availability during the current season (Uscola et al., 2015; Villar-Salvador et al.,

2015). Thus, the N increases observed in fertilized seedlings would be stored to promote growth during the second and subsequent years rather than used in growth during the current year.

It is possible that factors other than N were limiting growth in this experiment. Similar responses have been found in poplar plantations established on soils with high organic matter (Pinno and Bélanger, 2009) or pH (DesRochers et al., 2006) where an increase in available N resulted in very little growth benefit, likely because tree growth was not limited by N on these soils. An increase in N availability, the only nutrient applied in this experiment, can render the effects of growth limitation by other nutrients more apparent. For instance, Sloan and Jacobs (2013) reported much stronger growth responses after two growing seasons on this same site and for the same rate of N addition when a complete and balanced fertilizer (CRF or IAF) was used. Phosphorus is one of the most limiting nutrients in the boreal forest of Alberta (Strong and La Roi, 1985), and it is possible that P was more limiting than N in this study. Foliar P concentrations in this experiment ranged from 0.7 to 1.9 mg g⁻¹ (Table 2), below the critical threshold identified for aspen by Hansen et al. (1996). Other experiments involving young planted aspen in Alberta have shown that fertilization increased growth when N applications were accompanied with P additions (Pinno et al., 2012; van den Driessche et al., 2003). So, identifying this nutrient-specific differential growth response based on reclaimed soil properties should be a first important step in refining fertilizer prescriptions that better meet the demands of seedlings on a given reclamation site. Additionally, further research is warranted in order to better understand the manner in which seedling growth and different species-dependent patterns of biomass and N allocation may respond to the nutrient balance of fertilizer products applied on reclamation sites.

4.3. Recovery of fertilizer nitrogen

After the first growing season, the majority of recovered fertilizer-derived ¹⁵N remained in soil surrounding fertilized seedlings for both CRF and IAF (Fig. 4). This result is consistent with findings of prior experiments. For instance, Chang et al. (1996) found around 70% of applied ¹⁵N in the soil, and 80% of applied ¹⁵N was recovered from the soil by Preston et al. (1990). Generally, microbial immobilization is a major N sink in soils (Chang and Preston, 2000; Staples et al., 1999), and this N can remain unavailable for uptake by outplanted seedlings or early successional plants even eight years after application (Preston and Mead, 1994). The lower ¹⁵N recovery in the CRF treatment may be due to a significant amount of ¹⁵N still in the fertilizer prills. Nutrient release of many CRF products can continue over longer periods than a year (Fan et al., 2002; Jacobs et al., 2005). Hangs et al. (2003) found that the majority of applied N (around 58–73%) remained in the CRF prills even after the second growing season.

Relative proportions of applied ¹⁵N recovered from vegetation and seedlings in this study were generally lower than first-year CRF-derived ¹⁵N uptake rates observed by Hangs et al. (2003) for white spruce and jack pine seedlings and competing vegetation on a boreal planting site in Saskatchewan, Canada. Differences between studies may reflect variation in soil nutrient dynamics between the constructed soils of the current study and the intact native forest soils investigated by Hangs et al. (2003). However, values obtained were similar to those in other studies using broadcast applications where around 0.4–15% of applied ¹⁵N was found in seedlings after one (Clinton and Mead, 1994) or two (Staples et al., 1999) growing seasons.

Fertilizer N recovery has been studied using ¹⁵N tracer technique in several conifers with generally low N recoveries of 10–25% (Hulm and Killham, 1990; Preston et al., 1990). In our case, much of the unaccounted-for ¹⁵N, especially from the IAF treat-

ment, likely moved beyond the sampling area. Part of the ¹⁵N not recovered would be lost from the ecosystem by leaching or gaseous losses (Hulm and Killham, 1990; Preston et al., 1990). It is probable that the constructed soils, altered hydrology, and exposure inherent to mine reclamation sites would prove conducive to increased loss of applied N due to off-site movement (Chang et al., 1996). Depending on substrate additions (i.e. relative proportions of peat or forest floor material), the cation exchange capacity of substrates in these reclaimed areas may be low due to limited organic matter or clay content, and fertilization would likely supply nutrients at levels that exceed both the total exchange capacity of the substrate and the plant uptake rates, thus leading to off-site movement.

In this study, competing vegetation and seedlings contained similar amounts of fertilizer-derived ¹⁵N in both CRF and IAF treatments (Fig. 4), unlike other studies that have reported higher uptake by native early successional species than by planted seedlings. For example, Chang et al. (1996) found that the applied ¹⁵N recovered by seedlings was 4.1% in western redcedar (*Thuja plicata* Donn ex D. Don.), 2.0% in western hemlock (*Tsuga heterophylla* (Raf.) Sarg.), and 4.9% in Sitka spruce (*Picea sitchensis* (Bong.) Carrière) while competing vegetation recovered 19.6%, 31.8%, and 16.2% of applied ¹⁵N, respectively. Also, Preston et al. (1990) found that 22.0% of ¹⁵N was taken up by native early successional vegetation species while ¹⁵N uptake of outplanted lodgepole pine (*Pinus contorta* Dougl. var. *latifolia* Engelm.) and Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco) trees ranged from 0.4% to 10.1% of the applied fertilizer ¹⁵N. Low uptake of applied ¹⁵N by early successional vegetation in this study indicated that such species were not principal competitors for the applied fertilizer during the study period.

5. Conclusions

The majority of recovered applied ¹⁵N was in the soil, with competing vegetation and seedlings accounting for relatively small amounts of fertilizer-derived ¹⁵N. Seedlings absorbed slightly more ¹⁵N from fertilizer than did competing vegetation for both IAF and CRF. Despite low rates of fertilizer recovery for both CRF and IAF, both white spruce and trembling aspen seedlings showed increased diameter growth in response to CRF, as well as elevated N concentration in roots and foliage for both CRF and IAF seedlings. Foliar N content and ¹⁵N content of both CRF and IAF seedlings increased relative to controls, but root N content and ¹⁵N content were only elevated in the CRF treatment. For all measured parameters, seedlings of the CRF treatment performed as well as or better than IAF seedlings, despite the 80% lower rate of N applied to seedlings in the CRF treatment. Patterns of biomass allocation were shown to differ substantially between species, with white spruce allocating biomass preferentially to foliage and stems, while trembling aspen allocated biomass primarily to roots and stems. Species differences were observed with regard to allocation of total N and assimilated fertilizer-derived ¹⁵N, with white spruce allocating N preferentially to foliage, while trembling aspen balanced N allocation between roots and foliage. Findings suggest that a shift from broadcast IAF to targeted soil applications of CRF on reclamation areas similar to this study site would offer similar or improved first-season seedling outcomes compared to broadcast IAF while greatly reducing overall quantities of N applied during the early stages of forest restoration, much of which appears to be lost from the site of application regardless of fertilizer type.

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